

Linear Algebra

In-Depth Study Notes, Rebuilt from Your Lecture Notes

Today's Session: Chapter 1 — Matrices

How to Read These Notes

Your source pages came through as raw, unformatted text (matrices with no brackets, sums written as “!”, symbols dropped), which makes them painful to study from directly. These notes rebuild Chapter 1 (Matrices) properly: every definition and theorem restated precisely, in the correct order, with plain-English explanations, fresh worked examples (so nothing is just a copy of the source), and practice problems in the same style as the textbook's own exercises. Colour code used throughout:

- **Blue** = original *Definition*, restated precisely.
- **Purple** = original *Theorem / Result*.
- **Green** = “In Plain English” – the same idea in everyday language.
- **Grey** = a fully worked example (with fresh numbers).
- **Orange** = a practice problem for you to try.
- **Coloured bold text** = a jargon word, explained the first time it appears.

Only today's chapter, as requested. Message me next time and I'll build Chapter 2 (Linear System of Equations) in the same format.

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1 §1.1 Definition of a Matrix

A **matrix** is simply a rectangular table of numbers. Matrices are the central object of the whole course because they let us package an entire system of equations, or an entire geometric transformation, into a single object we can add, multiply, and manipulate with clean rules.

Definition 1.1.1 – Matrix

A rectangular array of numbers is called a **matrix**. The horizontal arrays are its **rows** and the vertical arrays are its **columns**. A matrix with m rows and n columns is said to have **order** $m \times n$ (read “ m by n ”), written

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix},$$

where a_{ij} (the entry in row i , column j) is called the **(i, j) -entry**. We often abbreviate this as $A = [a_{ij}]$, leaving the order understood from context. A matrix with only one column is a **column vector**; a matrix with only one row is a **row vector**.

In Plain English

Think of a matrix as a spreadsheet with a fixed number of rows and columns – no missing cells, every cell holds a number. “Order 2×3 ” just means 2 rows, 3 columns; note the row count always comes first. If $A = \begin{pmatrix} 1 & 3 & 7 \\ 4 & 5 & 6 \end{pmatrix}$, then $a_{11} = 1$, $a_{12} = 3$, $a_{13} = 7$, $a_{21} = 4$, $a_{22} = 5$, $a_{23} = 6$ – read the subscripts as “row, column”, in that order, every time.

Equality of Two Matrices

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ (both of order $m \times n$) are **equal** if $a_{ij} = b_{ij}$ for every i, j . In particular, matrices of different orders are never equal, no matter what numbers they contain.

Worked Example – a linear system as a matrix

The system of equations $3x + 4y = 10$, $x - 2y = 1$ can be packaged as the **augmented matrix**

$$\left[\begin{array}{cc|c} 3 & 4 & 10 \\ 1 & -2 & 1 \end{array} \right],$$

where the vertical line just reminds us where the “=” sign used to be. This is exactly the idea Chapter 2 (next session) builds an entire solution method around.

Special Matrices**Definition 1.1.5 – Special Matrices**

- (1) The **zero matrix** 0 has every entry equal to 0.
- (2) A matrix is **square** if its number of rows equals its number of columns; a square matrix of order $n \times n$ is just called “of order n ”.
- (3) In a square matrix $A = [a_{ij}]$ of order n , the entries $a_{11}, a_{22}, \dots, a_{nn}$ are the **diagonal entries**, forming the **principal diagonal**.
- (4) A square matrix is a **diagonal matrix** if $a_{ij} = 0$ whenever $i \neq j$ (every off-diagonal entry is 0); written $D = \text{diag}(d_1, \dots, d_n)$. If every d_i equals the same number d , D is called a **scalar matrix**.
- (5) The **identity matrix** I_n is the diagonal matrix with every diagonal entry equal to 1.
- (6) A square matrix $A = [a_{ij}]$ is **upper triangular** if $a_{ij} = 0$ for $i > j$ (everything strictly

below the diagonal is 0), and **lower triangular** if $a_{ij} = 0$ for $i < j$. Either case is called **triangular**.

In Plain English

These are just graphs of “where the zeros have to be”:

- *Diagonal*: everything off the main diagonal is 0 – only the diagonal entries can be nonzero.
- *Scalar matrix*: a diagonal matrix where every diagonal entry is the *same* number – e.g. $4I_3$. These behave almost like ordinary numbers when multiplied with other matrices (you’ll see why in §1.2).
- *Identity matrix* I_n : the “do nothing” matrix – multiplying by it (once we define multiplication) leaves everything unchanged, exactly like multiplying a number by 1.
- *Triangular*: picture the matrix cut by its diagonal – one whole triangular half is forced to be zero. Upper triangular = zeros in the bottom-left triangle; lower triangular = zeros in the top-right triangle.

Worked Example – classifying matrices

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \quad S = 4I_3 = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 1 & 4 \\ 0 & -3 & 1 \\ 0 & 0 & 5 \end{pmatrix}.$$

D is diagonal but *not* scalar (its diagonal entries 2, -1 , 5 aren’t all equal). S is a scalar matrix (and hence also diagonal, upper triangular, and lower triangular all at once – diagonal matrices satisfy every triangular condition trivially). U is upper triangular but not diagonal (the entries 1, 4, 1 above the diagonal are nonzero) and not lower triangular.

Practice Problems – §1.1

- 1.1.1 Write down the order of $A = \begin{pmatrix} 2 & 5 & 1 \\ 0 & -1 & 4 \end{pmatrix}$ and identify a_{11} , a_{23} .
- 1.1.2 Construct the augmented matrix for the system $2x - y + z = 4$, $x + 3y - z = 0$.
- 1.1.3 Give a 3×3 diagonal matrix that is *not* a scalar matrix, then give one that *is* a scalar matrix.
- 1.1.4 Is $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$ upper triangular, lower triangular, both, or neither? What about $\begin{pmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{pmatrix}$?
- 1.1.5 Write out I_4 in full.
- 1.1.6 (*Challenge*) Can a non-square matrix ever be called “diagonal” under Definition 1.1.5? Explain why the definition only makes sense for square matrices.

2 §1.2 Operations on Matrices

Transpose

Transpose of a Matrix

The **transpose** of an $m \times n$ matrix $A = [a_{ij}]$ is the $n \times m$ matrix $A^t = [b_{ij}]$ with $b_{ij} = a_{ji}$ – i.e. the rows of A become the columns of A^t (and vice versa).

Theorem 1.2.2

For any matrix A , $(A^t)^t = A$.

In Plain English

Transposing “flips” the matrix across its main diagonal. Flip it twice and you’re back where you started – exactly like flipping a pancake. A row vector’s transpose is a column vector, and vice versa.

Worked Example

$$A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \\ -1 & 4 \end{pmatrix} \text{ (order } 3 \times 2) \Rightarrow A^t = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 4 \end{pmatrix} \text{ (order } 2 \times 3).$$

Addition and Scalar Multiplication

Addition and Scalar Multiplication

For two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ of the *same* order, $A + B := [a_{ij} + b_{ij}]$ (addition is undefined if the orders differ). For $k \in \mathbb{R}$, $kA := [ka_{ij}]$ (**scalar multiplication**: scale every entry).

Theorem 1.2.5 – Properties of Addition

Let A, B, C be matrices of order $m \times n$ and $k, l \in \mathbb{R}$. Then

- (1) $A + B = B + A$ (**commutativity**);
- (2) $(A + B) + C = A + (B + C)$ (**associativity**);
- (3) $k(lA) = (kl)A$;
- (4) $(k + l)A = kA + lA$.

The zero matrix $0_{m \times n}$ is the **additive identity**: $A + 0 = 0 + A = A$. Every A has an **additive inverse** $-A := (-1)A$, with $A + (-A) = 0$.

In Plain English – why is this “obvious”?

Everything here is just ordinary real-number arithmetic applied one entry at a time. For instance, $A + B = B + A$ holds because each entry satisfies $a_{ij} + b_{ij} = b_{ij} + a_{ij}$ – real numbers commute, so the matrices built entry-by-entry from them commute too. Every other property in Theorem 1.2.5 is proved the exact same way: reduce a matrix statement to the corresponding statement about real numbers, which you already know is true.

Worked Example

$$A = \begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix}, B = \begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix}.$$

$$A + B = \begin{pmatrix} 5 & -1 \\ 2 & 2 \end{pmatrix}, \quad 3A - 2B = \begin{pmatrix} 3 & -6 \\ 9 & 0 \end{pmatrix} - \begin{pmatrix} 8 & 2 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} -5 & -8 \\ 11 & -4 \end{pmatrix}.$$

Matrix Multiplication – the Interesting Operation**Definition 1.2.8 – Matrix Product**

Let $A = [a_{ij}]$ be $m \times n$ and $B = [b_{ij}]$ be $n \times r$. Their **product** AB is the $m \times r$ matrix $C = [c_{ij}]$ with

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

AB is defined **only if** the number of columns of A equals the number of rows of B .

In Plain English

To get the (i, j) -entry of AB : walk along row i of A and down column j of B at the same time, multiplying corresponding entries and adding up the results (a “row times column” dot product). This is why the inner dimensions must match: row i of A has n numbers, column j of B must also have n numbers, or there’s nothing to pair them with.

Worked Example – computing a product

$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{pmatrix}$ (2×3), $B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 2 \end{pmatrix}$ (3×2). Since A has 3 columns and B has 3 rows, AB is defined and has order 2×2 :

$$c_{11} = 1(1) + 2(2) + 0(0) = 5, \quad c_{12} = 1(0) + 2(1) + 0(2) = 2,$$

$$c_{21} = 0(1) + 1(2) + 3(0) = 2, \quad c_{22} = 0(0) + 1(1) + 3(2) = 7.$$

$$AB = \begin{pmatrix} 5 & 2 \\ 2 & 7 \end{pmatrix}.$$

Note BA would be 3×3 (a completely different-sized object) – when both products exist, they need not even be comparable, let alone equal.

Definition 1.2.9 – Commuting Matrices

Two square matrices A, B of the same order **commute** if $AB = BA$.

Remark 1.2.10

For any square A of order n : $AI_n = I_nA = A$. Any scalar matrix dI_n commutes with *every* square matrix of order n . But in general, **matrix multiplication is not commutative**.

Worked Example – multiplication is not commutative

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

$$AB = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad BA = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

$AB \neq BA$ (compare the top-left entries: $1 \neq 2$). Order matters in matrix multiplication – always keep track of which side you’re multiplying on.

Theorem 1.2.11 – Properties of Multiplication

Whenever the products below are defined:

- (1) $(AB)C = A(BC)$ (associativity);
- (2) $(kA)B = k(AB) = A(kB)$ for $k \in \mathbb{R}$;
- (3) $A(B + C) = AB + AC$ (distributivity);
- (4) $AI_n = I_n A = A$ for square A of order n ;
- (5) For $D = \text{diag}(d_1, \dots, d_n)$ and square A of order n : the i -th row of DA is d_i times the i -th row of A ; the j -th column of AD is d_j times the j -th column of A .

In Plain English

Part (5) is a genuinely useful shortcut, worth memorising as a rule of thumb: **left-multiply by a diagonal matrix to rescale rows; right-multiply by a diagonal matrix to rescale columns**. This turns what looks like a full matrix multiplication into simple row/column scaling – much faster in practice, and a trick you’ll reuse constantly once you reach Gaussian elimination and diagonalisation.

Practice Problems – §1.2

- 1.2.1** For $A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \\ -1 & 4 \end{pmatrix}$, find A^t and state its order.
- 1.2.2** Let $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 4 \\ 2 & 0 \end{pmatrix}$. Compute AB and BA . Do they commute?
- 1.2.3** Let A be 2×3 and B be 3×5 . Explain, using Definition 1.2.8, why AB is defined but BA is not.
- 1.2.4** Compute DA and AD for $D = \text{diag}(2, -1)$ and $A = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}$, and check your answer against Theorem 1.2.11(5).
- 1.2.5** If $A + B = A$, show (directly from the definitions) that $B = 0$.
- 1.2.6** (*Challenge*) Find a nonzero 2×2 matrix A with $A^2 = 0$ but $A \neq 0$ (this shows matrices can behave very differently from ordinary numbers – no nonzero real number squares to 0).

3 §1.3 Some More Special Matrices

Definition 1.3.1 – Symmetric, Skew-Symmetric, Orthogonal

A square matrix A (real entries) is

- **symmetric** if $A^t = A$;
- **skew-symmetric** if $A^t = -A$;
- **orthogonal** if $AA^t = A^tA = I$.

In Plain English

Symmetric: the matrix looks the same after flipping across the diagonal – entry (i, j) always equals entry (j, i) . Skew-symmetric: flipping *negates* every entry, which forces every diagonal entry to be 0 (since $a_{ii} = -a_{ii}$ only when $a_{ii} = 0$). Orthogonal: multiplying A by its own transpose gives the identity – geometrically, these are exactly the matrices that represent rotations and reflections (they preserve lengths and angles).

Worked Example

$$S = \begin{pmatrix} 2 & -1 & 3 \\ -1 & 0 & 4 \\ 3 & 4 & 5 \end{pmatrix} \text{ (symmetric: } S^t = S)$$

$$K = \begin{pmatrix} 0 & 2 & -1 \\ -2 & 0 & 3 \\ 1 & -3 & 0 \end{pmatrix} \text{ (skew-symmetric: } K^t = -K)$$

A classic orthogonal matrix is the rotation matrix $R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$: a direct computation using $\cos^2 \theta + \sin^2 \theta = 1$ confirms $R(\theta)^t R(\theta) = I_2$ for every angle θ .

A Useful Fact (source: Exercise 1.3.3)

For *any* square matrix A , define $S = \frac{1}{2}(A + A^t)$ and $T = \frac{1}{2}(A - A^t)$. Then S is symmetric, T is skew-symmetric, and $A = S + T$. That is: **every square matrix splits uniquely into a symmetric part plus a skew-symmetric part.**

Worked Example

$$A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}, \text{ so } A^t = \begin{pmatrix} 3 & -1 \\ 1 & 2 \end{pmatrix}.$$

$$S = \frac{1}{2}(A + A^t) = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}, \quad T = \frac{1}{2}(A - A^t) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Check: S is symmetric, T is skew-symmetric, and $S + T = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} = A$. ✓

Nilpotent and Idempotent Matrices

A square matrix A is **nilpotent** if $A^k = 0$ for some positive integer k ; the least such k is the **order of nilpotency**. A is **idempotent** if $A^2 = A$.

In Plain English

Nilpotent matrices are “eventually zero” under repeated multiplication – strictly triangular matrices (zeros on *and* above/below the diagonal) are the standard example. Idempotent matrices are “stable” under multiplication – applying the transformation twice does nothing more than applying it once. This is exactly the algebraic signature of a **projection** (think: squashing 3D space flat onto a 2D plane – doing that twice is the same as doing it once).

Worked Example

Let $N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. Direct computation gives $N^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq 0$ but $N^3 = 0$. So N is nilpotent of order 3.

Let $P = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$. Direct computation gives $P^2 = P$, so P is idempotent – and notice it is *not* a diagonal matrix, so idempotence is a genuinely richer property than the trivial examples I and 0 .

Submatrices and Block Matrices**Submatrix**

A matrix obtained by deleting some rows and/or columns of a given matrix A is called a **submatrix** of A .

In Plain English

Just cross out whichever rows/columns you don’t want and see what’s left. This sounds trivial, but it becomes a genuinely powerful idea once we reach determinants (minors and cofactors, in your next session’s material, are built entirely out of submatrices) and rank.

Rather than treating a matrix as one undivided block of numbers, it is often useful to slice it into smaller **blocks** (sub-rectangles) and work with those blocks the way we’d work with ordinary entries. This pays off when a matrix has natural sub-structure (e.g. some blocks are zero or identity), and it is the standard tool used later for induction proofs on matrix size.

Theorem 1.3.6 – Block Multiplication

Suppose A (order $n \times m$) is split into blocks $A = [P \ Q]$, with P of order $n \times r$ and Q of order $n \times (m - r)$; and B (order $m \times p$) is split into blocks $B = \begin{pmatrix} H \\ K \end{pmatrix}$, with H of order $r \times p$ and K of order $(m - r) \times p$. Then

$$AB = PH + QK.$$

In Plain English

As long as you slice A into column-blocks and B into matching row-blocks (the block sizes have to line up, exactly like ordinary matrix multiplication), you can multiply block-by-block and add, and you get the *same* answer as multiplying the whole matrices out entry by entry. This is genuinely useful when some blocks are especially simple (zero or identity), since those blocks vanish or drop out of the sum entirely.

Worked Example – block multiplication

Let $A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$, split as $P = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Let $B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 1 \\ 1 & 0 \end{pmatrix}$, split as

$H = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $K = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$. Block multiplication gives

$$PH + QK = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}.$$

Multiplying A and B directly, entry by entry, gives the exact same $\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$ – confirming the theorem.

Bonus – the Trace

One more quantity worth knowing now, since it will resurface constantly: for a square matrix A of order n , the **trace** is $\text{tr}(A) := a_{11} + a_{22} + \cdots + a_{nn}$ (just add up the diagonal entries). Two handy facts: $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$, and (less obviously) $\text{tr}(AB) = \text{tr}(BA)$ even though AB and BA themselves are usually *not* equal.

Practice Problems – §1.3

- 1.3.1** Verify $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ is symmetric and $\begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$ is skew-symmetric, directly from the definitions.
- 1.3.2** Confirm $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ is nilpotent by computing A^2 and A^3 yourself (don't just trust the worked example above).
- 1.3.3** Find a 2×2 idempotent matrix that is diagonal but is not I_2 or 0 . (*Hint*: what diagonal entries d satisfy $d^2 = d$?)
- 1.3.4** For $A = \begin{pmatrix} 4 & 2 \\ 0 & -2 \end{pmatrix}$, compute $S = \frac{1}{2}(A + A^t)$ and $T = \frac{1}{2}(A - A^t)$, and check $S + T = A$.
- 1.3.5** Split a 4×4 matrix of your choice into four 2×2 blocks and verify Theorem 1.3.6 against a second matrix of your choice (check your block computation against direct multiplication, as in the worked example).
- 1.3.6** (*Challenge*) Prove that every diagonal entry of a skew-symmetric matrix must be 0, directly from $A^t = -A$.

4 §1.4 Matrices over Complex Numbers

Everything above still works when the entries are complex numbers, with one extra layer: since complex numbers have conjugates, matrices get a “conjugate transpose” operation that plays the same role transpose does for real matrices.

Definition 1.4.1 – Conjugate and Conjugate Transpose

Let $A = [a_{ij}]$ be a matrix over $\mathbb{C}$.

- (1) The **conjugate** of A , denoted $\overline{A}$, is $[\overline{a_{ij}}]$ (conjugate every entry).

- (2) The **conjugate transpose** (or **adjoint**) of A , denoted A^* , is the matrix $B = [b_{ij}]$ with $b_{ij} = \overline{a_{ji}}$ – i.e. transpose, then conjugate every entry (equivalently, $A^* = \overline{A}^t$).

Hermitian, Skew-Hermitian, Unitary, Normal

A square matrix A over $\mathbb{C}$ is called

- **Hermitian** if $A^* = A$;
- **skew-Hermitian** if $A^* = -A$;
- **unitary** if $A^*A = AA^* = I$;
- **Normal** if $AA^* = A^*A$.

Remark 1.4.2

If $A = [a_{ij}]$ has *real* entries, then $A^* = A^t$: conjugation does nothing to a real number, so the conjugate-transpose collapses back to the ordinary transpose. In other words, for real matrices, Hermitian = symmetric, skew-Hermitian = skew-symmetric, and unitary = orthogonal – the complex definitions above are the natural, exact generalisations of the real ones from §1.3.

In Plain English

Whenever you see a “*” instead of a “ t ”, mentally read it as “transpose, but also conjugate everything along the way.” Every fact you already believe about symmetric/skew-symmetric/orthogonal real matrices has a Hermitian/skew-Hermitian/unitary twin for complex matrices, with proofs that go through unchanged once you remember to conjugate.

Worked Example

Let $A = \begin{pmatrix} 1+i & 2 \\ 3-i & i \end{pmatrix}$. Then

$$\overline{A} = \begin{pmatrix} 1-i & 2 \\ 3+i & -i \end{pmatrix}, \quad A^* = \begin{pmatrix} 1-i & 3+i \\ 2 & -i \end{pmatrix}.$$

Now let $H = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}$. Computing H^* : the (1,1)-entry conjugates to 2, the (2,2)-entry conjugates to 3, and the off-diagonal entries $1-i$ and $1+i$ swap positions and conjugate into each other. So $H^* = H$: H is **Hermitian**. This is the general pattern for Hermitian matrices: real numbers on the diagonal, and mirror-image entries across the diagonal that are complex conjugates of each other.

Practice Problems – §1.4

- 1.4.1 For $A = \begin{pmatrix} 2-i & 4 \\ 0 & 3+i \end{pmatrix}$, find $\overline{A}$ and A^* .
- 1.4.2 Is $\begin{pmatrix} 5 & 2+3i \\ 2-3i & -1 \end{pmatrix}$ Hermitian? Justify using Definition 1.4.1.
- 1.4.3 Verify directly that I_n is unitary for any n .
- 1.4.4 Using $A^* = -A$, prove that every diagonal entry of a skew-Hermitian matrix must be *purely imaginary* (i.e. of the form bi for real b , possibly 0). (*Hint*: what does $a_{ii} = -\overline{a_{ii}}$ force, writing $a_{ii} = x + yi$?)

1.4.5 (Challenge) Show that if A is unitary, then for any two column vectors $u, v \in \mathbb{C}^n$, multiplying both by A does not change their inner product u^*v . (This is why unitary matrices are often described as “distance and angle preserving.”)

Session Wrap-Up

What You Now Know

- How to read off the order of a matrix and its entries, and when two matrices count as equal.
- The whole family of special matrices (zero, diagonal, scalar, identity, triangular) and how to recognise each by its pattern of zeros.
- Transpose, addition, and scalar multiplication – all just entrywise real-number arithmetic.
- Matrix multiplication: the row-times-column rule, why dimensions must match, and why order matters ($AB \neq BA$ in general).
- Symmetric/skew-symmetric/orthogonal matrices, the universal $A = S + T$ decomposition, nilpotent and idempotent matrices, and block multiplication.
- The complex analogues: conjugate transpose, and Hermitian/skew-Hermitian/unitary/normal matrices.

Self-Check Before Moving On

You should be able to, without looking anything up:

1. State the row-times-column rule for matrix multiplication, and explain why AB can be defined while BA isn't.
2. Split any square matrix into its symmetric and skew-symmetric parts.
3. Recognise a Hermitian matrix on sight (real diagonal, conjugate-mirrored off-diagonal entries).

If any of these feel shaky, revisit that box above before moving on – Chapter 2 (Linear System of Equations) leans on matrix multiplication and the identity matrix constantly, right from its first page.

That's it for today. Message me next time and I'll build Chapter 2 – Linear System of Equations (row operations, Gauss elimination, rank, determinants) – in the same format.